

DIG Northern Starter Kit

 **Version 8.0**

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For transcription-labeling of RNA with digoxigenin and SP6/T7/T3 RNA polymerases and chemiluminescent detection with CDP-Star, ready-to-use.

Cat. No. 12 039 672 910

Kit for 10 labeling reactions and
detection of 10 blots of $10 \times 10 \text{ cm}^2$

Store the kit at -15 to -25°C

1. Preface

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1.2 Kit contents

Bottle/ Cap	Label	Content including function
1a	Labeling Mix	• 40 μl • 5 $\times$ conc. labeling mixture containing optimal concentrations of unlabeled nucleotides and DIG -11-UTP • Clear viscous solution • For efficient <i>in vitro</i> transcription of linearized template DNA
1b	Transcription Buffer	• 40 μl • 5 $\times$ conc. transcription buffer • Clear solution • For efficient <i>in vitro</i> transcription of linearized template DNA
2	SP6 RNA Polymerase	• 20 μl • [20 U/μl] • Synthesize RNA from a DNA template
3	T7 RNA Polymerase	• 20 μl • [20 U/μl] • Synthesize RNA from a DNA template
4	T3 RNA Polymerase	• 20 μl • [20 U/μl] • Synthesize RNA from a DNA template
5	Anti-digoxigenin-AP, Fab fragments	• 60 μl • [750 U/ml] • Polyclonal sheep anti-digoxigenin, Fab-fragments, conjugated to alkaline phosphatase • Clear solution
6	DNase I, RNase free	• 20 μl • [10 U/μl] • Degrades DNA template after transcription reaction
7	CDP- <i>Star</i> ready-to-use	• 20 ml • Chemiluminescent substrate for alkaline phosphatase
8	Actin RNA probe, DIG-labeled	• Antisense probe, length 588 bases • [10 ng/μl] • Standard for the quantification of DIG-labeled RNA and as a hybridization control
9	DIG Easy Hyb Granules	• 2 bottles for 100 ml solution each • For the hybridization of the DIG-labeled RNA probe
10	Blocking solution	• 2 $\times$ 100 ml • 10 $\times$ conc. • Yellow, viscous solution • For chemiluminescent detection procedure

Additional equipment and reagents required

In addition to the reagents listed above, you have to prepare several solutions, exclusively with DMPC or DEPC treated water. In the table you will find an overview about the equipment which is needed for the different procedures.

Detailed information is given in front of each procedure.

Procedure	Equipment	Reagents
3.3 DNA Template Preparation using 3.3.1 standard method or 3.3.1 RT-PCR and PCR	Thermocycler	• High Pure Plasmid Isolation Kit* • Phenol/Chloroform • Expand High Fidelity PCR System (incl. buffer)* • Primer (sense, antisense) • Nucleotides (dATP, dCTP, dGTP, dTTP, each 10 mM)
3.4 Procedure for DIG-RNA labeling	• Water-bath or heating block • Ice/water	• H₂O, sterile, double distilled • EDTA, 0.2 M, pH 8.0
3.5 Determination of labeling efficiency	• Nylon membranes positively charged* • UV-transilluminator or • UV-cross linker or • Oven (120°C or 80°C)	• RNA dilution buffer • DIG Wash and Block Buffer Set* or • Washing buffer • Maleic acid buffer • Detection buffer
3.6 Suitable gel system for Northern blots 3.6.1 Formaldehyde gels	• Gel equipment • Heatable water-bath (65° C) or • Heating block (65° C) • Ice	• 10 × MOPS, pH 7,0 (with NaOH) • Loading buffer • Gel solution • Running buffer
3.7 RNA blotting and fixation	• Nylon membranes positively charged* • Whatmann 3MM paper • UV- transilluminator or • UV-cross linker or • Oven (120°C or 80°C)	Depending on used method: • 20 × SSC or • 2 × SSC
3.8 Hybridization	• Nylon membranes, positively charged* • Ice/water • Water-bath • Shaking water-bath • or hybridization oven • Hybridization bags* or • Temperature resistant, sealable plastic or glass boxes, petri dishes or roller bottles Note: Do not use open trays or boxes with DIG Easy Hyb.	For stringency washes: • 2 × SSC, 0.1% SDS • 0.1 × SSC, 0.1% SDS
3.9 Immunological detection	• Hybridization bags* or • Development folders	• DIG Wash and Block Buffer Set* or • Washing buffer • Maleic acid buffer • Detection buffer
3.10 Stripping and reprobing of RNA blots	• Water bath • Hybridization bags*	• Double dist. water • Stripping buffer • 2 × SSC

2. Introduction

2.1 Product overview

Test principle

The DIG Northern Starter Kit generates DIG labeled, single stranded RNA probes of defined length by *in vitro* transcription of template DNA in the presence of digoxigenin-UTP, using SP6, T7 or T3 RNA polymerases.

Stage	Description
RNA labeling	DIG-labeled RNA probes are generated according to the <i>in vitro</i> transcription labeling technique. The labeling mix contains optimal concentrations of nucleotides and DIG-11-UTP and a specially developed transcription buffer are combined with the linearized DNA template and the appropriate RNA polymerase.
Hybridization	DIG-labeled RNA probes are used for hybridization to membrane blotted nucleic acids according to standard methods.
Immunological detection	The hybridized probes are immunodetected with anti-digoxigenin-AP, Fab fragments and are then visualized with the chemiluminescence substrate CDP- <i>Star</i> , ready-to-use. Enzymatic dephosphorylation of CDP- <i>Star</i> , ready-to-use by alkaline phosphatase leads to a light emission at a maximum wavelength of 465 nm (see fig. 1) which is recorded with the imaging device or on X-ray films. Exposure times are in the range of 5 to 30 min, only.

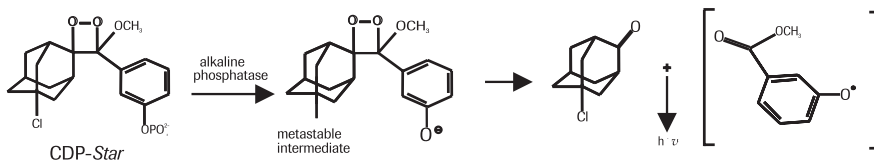


Fig. 1: Reaction of CDP-Star

Application

Northern blots on nylon membranes

Sample material

- Linearized plasmid, including the appropriate RNA polymerase promoter sequence (SP6, T3, T7).

Note: The length of the region to be transcribed should be in the range of 200 to 1000 bp. To avoid RNase contamination the DNA must be phenolized.

- Specially prepared PCR product (see chapter 3.3.2)

Assay time

This table lists the reaction time of the single steps

Step	Reaction time
RNA labeling	1 h 20 min
Hybridization	6 h or overnight
Immunological detection	1 h 40 min
Chemiluminescent signal detection	5-30 min

Number of tests

1 kit is sufficient for

- 10 labeling reactions of 1 µg DNA, yielding approx. 20 µg labeled RNA, each and detection of
- 10 blots of 10 × 10 cm².

**Kit storage/
stability**

The unopened kit is stable at –15 to –25° C until the expiration date printed on the label.

Once opened, please refer to the following table for appropriate storage

Kit component	vial	Storage/Stability
Labeling Mix and Transcription Buffer	1a	Stable at –15 to –25° C Note: Repeated freezing and thawing should be avoided. To avoid contamination we recommend to aliquot the solutions and to store in 2-3 portions.
	1b	
RNA polymerases	2-4	Stable at –15 to –25° C
Anti-digoxigenin-AP, Fab fragments	5	Stable at +2 to +8° C Note: Do not freeze!
CDP- <i>Star</i> , ready-to-use	7	Stable at +2 to +8° C Note: Store protected from light!
DIG Easy Hyb Granules	9	- Stable at +15 to +25° C - Once dissolved, the solution is stable for 1 month when kept sterile.
Blocking solution (10x concentrated)	10	Stable at +2 to +8° C Note: We recommend to store the concentrated blocking solution in aliquots at -15°C to -25°C.

**Sensitivity and
specificity**

Rare mRNAs can be detected in 0.1 µg of total RNA

Advantages

This table describes benefits and features of the kit:

Benefit	Feature
Accurate and fast	The use of the application tested and optimized kit components minimizes the hands-on-time required to label RNA probes and increases efficiency and reproducibility.
Sensitive	Rare transcripts can be detected in total human or plant RNA.
Time-saving	DIG-labeled RNA probes can be stored at –15 to –25° C in ethanol for at least one year. We recommend to freeze the probe in aliquots, and avoid repeated freeze/thaw cycles.

3. Procedures and required materials

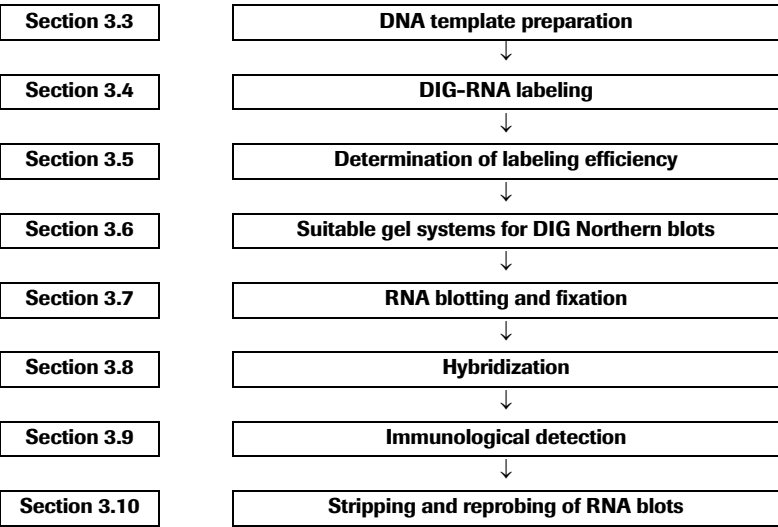
3.1 Before you begin

General handling recommendations

This table describes general hints for DIG labeling, detection and handling of RNA.

Recommendation	Guideline
Work under clean and RNase-free conditions	• Use DMPC or DEPC treated H₂O for preparation of all solutions • Autoclave solutions • Tween 20 should be added to previously sterilized solutions
Use clean incubation trays	Rigorously clean and rinse laboratory trays with RNase-ZAP (Sigma R-2020) or bake glass trays 8 hours at 200° C before use
Membrane handling requirements	• Wear powder-free gloves • Handle membrane only on the edges with clean forceps

3.2 Flow chart



3.3 DNA Template Preparation

3.3.1 Standard method

General information

- Plasmid DNA should be purified using the High Pure Plasmid Isolation Kit*.
- DNA template must be linearized at a restriction site downstream of the cloned insert. The sequence to be transcribed should be 200 to 1000 bp in length.
- To avoid transcription of undesirable sequences, use a restriction enzyme that leaves 5'overhang or blunt ends.
- After restriction digest, purify the DNA by phenol/chloroform extraction, followed by ethanol precipitation.
- Dissolve template DNA in 10 mM Tris, 0,1 mM EDTA, pH 8,0

Note: Too much EDTA inhibits the transcription reaction, do not use more than 0.1 mM EDTA in the TE buffer for dissolving the DNA.

3.3.2 Preparing DNA Template from total RNA using RT-PCR and PCR

Overview

The following table describes an alternative method for generating templates for *in vitro* transcription labeling of RNA with DIG without cloning.

Stage	Description
1	Prepare total RNA with one of the standard methods.
2	Run a RT-PCR with oligo(dT) Primer using the Expand Reverse Transcriptase System* .
3	Run the PCR with specially designed primers, including the sequence of the appropriate RNA polymerase promoter. Use standard PCR conditions; we recommend to use Expand High Fidelity PCR System for best results.

RNA polymerase promoter sequences

Please find in the following table the promoter sequences for the different RNA polymerases.

Promoter	Sequence
SP6 RNA polymerase	Note: The use of SP6 promotor consensus sequences cannot be recommended for PCR generated DNA fragments because experiments have shown that SP6 Polymerase can only initiate efficient transcription if the promoter sequences lies within a plasmid environment (3).
T7 RNA polymerase	5'TAATACGACTCACTATAGGG/X-mer or 5'TAATACGACTCACTATAGGA/X-mer
T3 RNA polymerase	5'AATTAACCCTCACTAAAGGG/X-mer

Additional equipment and reagents required

- Thermocycler
- Expand High Fidelity PCR System, including buffer
- Primer (sense, antisense)
- Nucleotides (dATP, dCTP, dGTP, dTTP, each 10 mM)

Procedure

Find in the following table the recommended protocol using the Expand High Fidelity PCR System*.

Step	Action																											
1	• Add the following reagents to a sterile, RNase-free reaction tube (on ice) in the following order:<table><tr><th>Reagent</th><th>Volume</th><th>Final conc.</th></tr><tr><td>sterile double dist. water, DMPC or DEPC-treated</td><td>variable</td><td>-</td></tr><tr><td>Expand buffer (10 ×) containing MgCl₂ (supplied with the enzyme)</td><td>5 μl</td><td>1.5 mM MgCl₂ (1 ×)</td></tr><tr><td>10 mM dATP, dCTP, dGTP, dTTP</td><td>each 1 μl</td><td>0.2 mM</td></tr><tr><td>Primer 1 (sense)</td><td>variable</td><td>300 nM</td></tr><tr><td>Primer 2 (anti-sense)</td><td>variable</td><td>300 nM</td></tr><tr><td>Expand High Fidelity</td><td>0.75 μl</td><td>2.6 U</td></tr><tr><td>cDNA</td><td>2 μl</td><td>-</td></tr><tr><td>Final volume</td><td>50 μl</td><td></td></tr></table> • Mix gently and centrifuge briefly.	Reagent	Volume	Final conc.	sterile double dist. water, DMPC or DEPC-treated	variable	-	Expand buffer (10 ×) containing MgCl ₂ (supplied with the enzyme)	5 μl	1.5 mM MgCl ₂ (1 ×)	10 mM dATP, dCTP, dGTP, dTTP	each 1 μl	0.2 mM	Primer 1 (sense)	variable	300 nM	Primer 2 (anti-sense)	variable	300 nM	Expand High Fidelity	0.75 μl	2.6 U	cDNA	2 μl	-	Final volume	50 μl	
Reagent	Volume	Final conc.																										
sterile double dist. water, DMPC or DEPC-treated	variable	-																										
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Expand High Fidelity	0.75 μl	2.6 U																										
cDNA	2 μl	-																										
Final volume	50 μl																											
2	Cycle 30 × as follows: <table><tr><th>Step</th><th>Temperature</th><th>Time</th></tr><tr><td>Denaturation</td><td>94°C</td><td>45 s</td></tr><tr><td>Annealing</td><td>60°C</td><td>45 s</td></tr><tr><td>Elongation</td><td>72°C</td><td>90 s</td></tr></table>	Step	Temperature	Time	Denaturation	94°C	45 s	Annealing	60°C	45 s	Elongation	72°C	90 s															
Step	Temperature	Time																										
Denaturation	94°C	45 s																										
Annealing	60°C	45 s																										
Elongation	72°C	90 s																										
3	Use your PCR product directly for labeling (Section 3.4).																											

3.4 DIG-RNA Labeling

Introduction RNA is labeled in an *in vitro* transcription reaction with digoxigenin-11-UTP using a Labeling mixture and an optimized Transcription Buffer.

Additional equipment and reagents required

- Water bath for 42° C and 37° C or heating block
- Ice/water

This table lists composition, storage and use of the additionally required reagents.

Solution	Composition	Storage/ Stability	Use
Water	Autoclaved, DMPC or DEPC treated double distilled water	+15 to +25° C, stable	Dilution of solutions
EDTA	0.2 M ethylenediamino-tetracetic acid, pH 8.0	+15 to +25° C, stable	Stopping the reaction

Procedure This procedure is designed for 1 µg of DNA template. Larger amounts can be labeled by scaling up of all components and volumes.

Step	Action
1	Add 1 µg linearized plasmid DNA or 4 µl PCR product (100-200 ng) and sterile RNase-free, DMPC or DEPC treated, double dist. water to a final volume of 10 µl to a sterilized reaction vial.
2	Reagent Labeling mix, 5 × (vial 1a) Volume 4 µl Transcription buffer 5 × (vial 1b) 4 µl RNA polymerase (SP6, T7 or T3) 2 µl • Add the following on ice: • Mix and centrifuge briefly. • Incubate for 1 h at 42° C.
3	• Add 2 µl DNase I, RNase-free to remove template DNA. • Incubate for 15 min at 37° C.
4	Stop the reaction by adding 2 µl 0.2 M EDTA (pH 8.0).

Labeling efficiency and size of labeled RNA

- In the standard reaction with 1 µg DNA per assay 67% of the nucleotides are incorporated into approx. 20 µg of newly synthesized DIG-labeled RNA within 1 h.
- The size of the labeled RNA is in the range of 200-1000 bases.

3.5 Determination of labeling efficiency

Introduction Determination of the yield of DIG-labeled RNA is very important for optimal and reproducible hybridization results. Too high probe concentrations in the hybridization step causes background, while too low concentrations leads to weak signals.

Test principle The preferential method for determination of labeling efficiency of probes is the direct detection method in comparison to the control RNA (vial 8).

Stage	Description
1	• A series of dilutions of DIG-labeled RNA is applied to a small strip of nylon membrane, positively charged*. • Part of the nylon membrane is preloaded with defined dilutions of control RNA, used as standards.
2	• The nylon membrane is subjected to immunological detection with anti-digoxigenin-AP and CDP-<i>Star</i> ready-to-use. • The intensities of the dilution series of DIG-labeled RNA and control RNA are compared by exposure to imaging device or X-ray film.

Additional equipment required

- Nylon membranes, positively charged*
- UV-transilluminator or
- UV-cross linker or
- Oven (120°C or 80°C)

Preparation of additional solutions required The Washing buffer, Maleic acid buffer, Blocking solution, and Detection buffer are also available in a ready-to-use form in the DIG Wash and Block Buffer Set DNase- and RNase-free*. These solutions are also used in the detection procedure of Chapter 3.9 and can be prepared in larger amounts.

Solution	Composition / Preparation	Storage/ Stability	Use
RNA dilution buffer	Mix DMPC treated double dist. water: 20 × SSC: Formaldehyd (37%) in the ratio 5 : 3 : 2	prepare fresh	Dilution of RNA
Washing buffer	0.1 M Maleic acid, 0.15 M NaCl; pH 7.5; 0.3% (v/v) Tween 20	+15 to +25° C, stable	Removal of unspecific bound antibody
Maleic acid buffer	0.1 M Maleic acid, 0.15 M NaCl; adjust with NaOH (solid) to pH 7.5	+15 to +25° C, stable	Dilution of Blocking solution
Detection buffer	0.1 M Tris-HCl, 0.1 M NaCl, pH 9.5 (20° C)	+15 to +25° C, stable	Adjustment of pH to 9.5

Preparation of kit working solutions

Please refer to the following table for the preparation of the working solutions.

Solution	Composition / Preparation	Storage/ Stability	Use
Blocking solution	Prepare a 1 × working solution by diluting the 10 × Blocking solution (bottle 10) 1:10 in Maleic acid buffer.	Always prepare fresh	Blocking of un-specific binding sites on the membrane
Antibody solution	Centrifuge anti-digoxigenin-AP (vial 5) for 5 min at 10 000 rpm in the original vial prior to each use, and pipet the necessary amount carefully from the surface. Dilute anti-digoxigenin-AP 1:10, 000 (75 mU/ml) in Blocking solution.	+2 to +8° C for 2 h	Binding to the DIG-label

Dilution series

Prepare a dilution series of your labeled probe and your control RNA as described in the table, where tube 1 is either a dilution of your labeling reaction to 10 ng/μl (expected yield of a standard labeling reaction is 20 μg of labeled RNA) or the control RNA (vial 8) which also has the concentration of 10 ng/μl.

Tube	RNA (μl)	From tube #	RNA Dilution Buffer (μl)	Dilution	Final concentration
1	-	diluted labeling reaction or vial 8	-	-	10 ng/μl
2	2	1	18	1:10	1 ng/μl
3	2	2	198	1:100	10 pg/μl
4	15	3	35	1:3.3	3 pg/μl
5	5	3	45	1:10	1 pg/μl
6	5	4	45	1:10	0.3 pg/μl
7	5	5	45	1:10	0.1 pg/μl
8	5	6	45	1:10	0.03 pg/μl
9	5	7	45	1:10	0.01 pg/μl
10	0	-	50	-	0

Procedure

The following table describes the direct detection.

Note: The volumes suggested below are for small membrane stripes of approx. 3 × 5 cm, processed in a small plastic container, such as a petri dish. Use sufficient buffer volumes to cover the membrane completely during all steps.

Step	Action
1	Apply 1 µl spots of tubes 3-10 from your labeled probes and the labeled control to the nylon membrane.
2	Fix the nucleic acid to the membrane by cross linking with UV-light or baking for 30 min at 120° C
3	- Transfer the membrane into a plastic container with 20 ml Washing buffer. - Incubate under shaking for 2 min at 15-25° C.
4	Incubate for 30 min in 10 ml Blocking solution .
5	Incubate for 30 min in 10 ml Antibody solution .
6	Wash with 20 ml Washing buffer , 2 x 15 min.
7	Equilibrate 2-5 min in 10 ml Detection buffer .
8	- Place membrane (with RNA side facing up) on a development folder (or hybridization bag) and apply CDP-Star approx. 4 drops (from the drop-per bottle 7) to the membrane. Immediately cover the membrane with the second sheet of the folder to spread the substrate evenly and without airbubbles over the membrane. Incubate for 5 min at + 15 to +25° C.
9	Seal the edges of the development folder around the damp membrane. Note: Drying of the membrane during exposure will result in dark background.
10	Expose to imaging device or to X-ray film for 5-25 min at +15 to +25° C.

Analysing the result

Compare the intensity of the spots out of your labeling reaction to the control and calculate the amount of DIG-labeled RNA.

3.6 Gel System suitable for DIG Northern Blots

General

- Standard protocols for gel electrophoresis can be used as described in Sambrook et al. [2].
- Gels lacking ethidium bromide are preferential used, because ethidium bromide can cause uneven background problems.
- We recommend formaldehyde denaturation and MOPS/formaldehyde gels, because to our experience they are easy to handle and reliable.

Target amounts

If you are using....	Then....
total RNA	load maximally 1 µg per lane
mRNA	load 100 ng per lane

3.6.1 Formaldehyde Gel

Solutions required The following buffers are required for running a formaldehyde gel:

Solution	Composition
10 × MOPS, pH 7.0 (with NaOH)	200 mM MOPS 50 mM NaAc 20 mM EDTA
Loading buffer (always prepare fresh)	250 µl of 100% Formamide 83 µl of 37% Formaldehyde 50 µl 10 × MOPS 50 µl 100% Glycerol 10 µl 2.5% Bromphenolblue 57 µl DEPC/DMPC treated water
Gel with Formaldehyd (2%)	1.8 g Agarose 141.9 ml 1 × MOPS 8.1 ml 37% Formaldehyde
Running buffer	1 × MOPS

Preparation of samples

Find the following table the preparation of the samples

Step	Action
1	Add 20 µl (or 2-3 volume) of Loading buffer to the RNA probe.
2	Denature the RNA probe/Loading buffer mix at 65°C for 10 min.
3	Chill the RNA probe/Loading buffer mix on ice for 1 min.
4	Run the gel wit 3-4 V/cm in RNase free gel boxes for at least 2 h (preferably over-night) until the RNAs are well separated..
5	To access the quality of the target RNA after electrophoresis, stain the gel briefly in 0.25 - 0.5 µg/ml ethidium bromide and examine the gel under UV light.

3.7 RNA transfer and fixation

Transfer methods and membranes

- When using formaldehyde gels, please rinse gels prior to blotting for 2 × 15 min in 20 × SSC
- All common types of RNA transfer methods are suitable for subsequent DIG hybridization [7]
- Best results are obtained when gels are blotted by capillary transfer with 20 × SSC overnight or at least for 6 hours
- Use only nylon membranes, positively charged

Note: We do not recommend to use alkali transfer (e.g., in 0.4 M NaOH) for RNA and DIG-labeled molecular weight markers*.

Fixation procedure

Fix the RNA to the membrane by any of the following procedures.

Note: Use sterile and RNase-free solutions and equipment

IF you want to...	THEN...
UV-crosslinking	• place the membrane on Whatman 3MM-paper soaked with 2 × SSC. • UV-crosslink the wet membrane without prior washing. • after the UV-crosslinking, rinse the membrane briefly in double distilled water and allow to air-dry.
bake at 120° C	• rinse membrane 2 × briefly in 2 × SSC. • bake the nylon membrane at 120° C for 30 min or according to the manufacturer's instructions.
bake at 80° C	• rinse membrane 2 × briefly in 2 × SSC • bake at 80° C for 2 h

Storage of the membrane

Find in the following table the storage conditions for the membrane.

IF...	THEN...
you want to go ahead.	use the membrane immediately for prehybridization.
you want to work later on.	store the membrane dry at +2 to +8 °C.

3.8 Hybridization

Factors which influence stringency

- The degree of homology of the probe to the target RNA (see "hybridization temperature") is an important factor for determining the appropriate conditions for hybridization.
- Stringency is influenced by temperature: high temperature increases stringency of hybridization, low temperature decreases stringency.
- For RNA : RNA hybridizations with DIG Easy Hyb, we recommend generally a hybridization temperature of 68° C. The temperature may have to be adjusted depending on the GC content and homology of probe to target.

Additional equipment required

- Nylon membranes, positively charged *
- Ice-water
- Water-bath
- Shaking water-bath
- or Hybridization oven
- Hybridization bags* or
- Temperature resistant plastic or glass boxes, petri dishes, roller bottles or sealable plastic bags.

Note: Do not use open trays with DIG Easy Hyb buffer.

Additional solutions required

- 20 × SSC
- 10% SDS

Preparation of kit working solution

Please refer to the following table to find the preparation protocol for the kit working solution.

Note: Use sterile, RNase-free solutions and equipment!

Solution	Composition / Preparation	Storage/ Stability	Use
DIG Easy Hyb Granules	Reconstitute granules (bottle 9) by carefully adding 64 ml sterile double distilled, DEPC or DMPC treated water in two portions to the plastic bottle, dissolve by stirring at 37° C. Note: DIG Easy Hyb Granules must be dissolved under RNase-free conditions.	at +15 to +25° C for 1 month	As prehybridization and hybridization solution

Procedure

Please refer to the following table.

Note: Hybridization temperature for Northern Blots with DIG Easy Hyb is 68 °C (for 100% homology of the probe to the target sequence), the DIG Easy Hyb working solution used for prehybridization has to be prewarmed to 68 °C.

Step	Action
1	• Prewarm an appropriate volume of DIG Easy Hyb (10–15 ml/100 cm² filter) to hybridization temperature (68 °C). • Prehybridize membrane with DIG Easy Hyb for 30 min with gentle agitation in an appropriate container. Note: Membranes should move freely, especially if you use several membranes in the same prehybridization solution.
2	Denature DIG-labeled RNA probe by boiling for 5 min and rapidly cooling in ice/water. Note: RNA is degraded by alkali-solutions!
3	Add denatured DIG-labeled RNA probe (100 ng/ml) to prewarmed DIG Easy Hyb (3.5 ml/100 cm ² membrane) and mix well but avoid foaming (bubbles may lead to background).
4	• Pour off prehybridization solution and add probe/hybridization mixture to membrane. • Incubate for 6 h or O/N at 68 °C with gentle agitation.

Storage of hybridization solution

In principle, DIG Easy Hyb containing DIG-labeled probe can be stored at –15 to –25° C and be reused when freshly denatured at 65°C for 10 min before use. We do not recommend reuse of DIG Easy Hyb containing RNA probes, because successful reuse depends on inherent probe stability and RNase-free working conditions.

Stringency washes

Please refer to the following table to find the procedure for stringency washes.

Step	Action
1	Wash 2 × 5 minutes in 2 × SSC , 0.1% SDS at +15 to +25 °C under constant agitation.
2	Wash 2 × 15 minutes in 0.1 × SSC, 0.1% SDS (prewarmed to wash temperature) at 68° C under constant agitation.

Note: The stringency of the final wash must be determined empirically. Depending on length and homology of the probe it will be necessary to adjust salt concentration. Fully homologous probes will often require 0.1x wash solution.

3.9 Immunological detection

General information

- Work under sterile, RNase-free conditions
- If the membrane is to be reprobed, do not allow the membrane to dry at any time.
- Use sufficient volume of all solutions.
- Make sure that membranes do not stick to trays during detection. Avoid that membranes stick together.
- Shake membranes during the whole procedure.
- When using laboratory trays for the detection procedure, they should be rigorously cleaned before use.
- Work under clean conditions when handling the chemiluminescent substrate solution and avoid phosphatase contamination.
- For chemiluminescent detection, membranes must also be sealed in plastic bags or development folders
- Do not use "Saran Wrap".

Additional reagents required

The Washing buffer, Maleic acid buffer, Blocking solution, and Detection buffer are also available in a ready-to-use form in the DIG Wash and Block Buffer Set, DNase and RNase free*.

Solution	Composition / Preparation	Storage/ Stability	Use
Washing buffer	0.1 M Maleic acid, 0.15 M NaCl; pH 7.5 (20° C); 0.3% (v/v) Tween 20	+15 to +25° C, stable	Removal of unspecific bound antibody
Maleic acid buffer	0.1 M Maleic acid, 0.15 M NaCl; adjust with NaOH (solid) to pH 7.5 (20° C)	+15 to +25° C, stable	Dilution of Blocking solution
Detection buffer	0.1 M Tris-HCl, 0.1 M NaCl, adjustment of pH to 9.5 (20° C)	+15 to +25° C, stable	Adjustment of pH to 9.5

Preparation of kit working solutions

The antibody solution should have been prepared for the Semi-quantitative determination of labeling (section 3.4) otherwise find in the following table the description for the preparation of the antibody working solution.

Solution	Composition / Preparation	Storage/ Stability	Use
Blocking solution	Prepare a 1 × working solution by diluting 10 × Blocking solution (bottle 10) 1:10 with Maleic acid buffer.	Always prepare fresh	Blocking of unspecific binding sites on the membrane
Antibody solution	Centrifuge anti-digoxigenin-AP (vial 5) for 5 min at 10 000 rpm in the original vial prior to each use, and pipet the necessary amount carefully from the surface. Dilute anti-digoxigenin-AP 1:10 000 (75 mU/ml) in Blocking solution.	2 h at +2 to +8° C	Binding to the DIG-labeled probe

Procedure

This table describes how to perform the immunological detection on a 100 cm² membrane in a suitable, not much larger tray.

Note: All incubations should be performed at +15 to +25° C with agitation. If the membrane is to be reprobed, do not allow the membrane to dry at any time.

Step	Action
1	After hybridization and stringency washes, rinse membrane briefly (1-5) min in Washing buffer .
2	Incubate for 30 min in 100 ml Blocking solution .
3	Incubate for 30 min in 50 ml Antibody solution .
4	Wash 2 × 15 min in 100 ml Washing buffer .
5	Equilibrate 2-5 min in 100 ml Detection buffer .
6	- Place membrane (with RNA side facing up) on a development folder (or hybridization bag) and quickly apply approx. 1 ml. CDP-Star, ready-to-use solution out of the dropper bottle (bottle 7) until the membrane is evenly soaked. Immediately cover the membrane with the second sheet of the folder to spread the substrate evenly and without airbubbles over the membrane. - Incubate for 5 min at +15 to +25° C.
7	Squeeze out excess liquid and seal the edges of the development folder around the damp membrane. Note: Drying of the membrane during exposure will result in dark background.
8	Expose to imaging device for 5-20 min or to X-ray film for 15-25 min at +15 to +25° C. Note: Luminescence continues for at least 24 hours. The signal increases in the first few hours. Multiple exposures can be taken to achieve the desired signal strength.

3.10 Stripping and reprobing of RNA blots

Before you begin Only membranes, which did not dry at any time during the hybridization and detection procedure, can be stripped.

Additional equipment required

- Hybridization bags
- Water-bath

Additional reagents required In the following table find the additional reagents required for the stripping of membranes.

Note: Use sterile, RNase free solutions.

Solution	Composition/Preparation	Storage/Stability	Use
H ₂ O	Sterile DMPC or DEPC treated double distilled water	+15 to +25° C, stable	Washing the membrane
Stripping buffer	50% Formamide 50 mM Tris/HCl 7,5 5% SDS	always prepare fresh	Removing DIG-labeled probe
20 × SSC stock solution	3 M NaCl 300 mM sodium citrate, pH 7.0 (Cat. No. 1666 681)	+15 to +25° C, stable	Stock solution for the preparation of 2 × SSC
2 × SSC	Dilute 20 × SSC stock solution 1:10 with double distilled, DEPC or DMDC-treated water.	+15 to +25° C, stable	Washing and storage of membrane

Procedure This table describes how to perform the stripping procedure.

Step	Action
1	Rinse membrane thoroughly in sterile DMPC or DEPC treated double distilled water .
2	Incubate 2 × 60 min at 80° C in Stripping buffer to remove the DIG-labeled probe.
3	Rinse 2 × 5 min in 2 × SSC .
4	Prehybridize and hybridize with a second probe or store membrane in a sealed bag.

Storage of stripped membrane Once the membrane is stripped, storage in Maleic acid buffer or 2 × SSC at +2 to +8°C is possible.

Make sure that the membrane does not dry!

4. Results

4.1 Typical results

Fig. 2

This figure shows the detection of the TSH receptor on Thyroid tissue with Graves disease in a Northern Blot.

Probe	DIG-labeled TSH receptor (100 ng/ml)
Template	Lane 1: 0.1 µg Thyroid RNA Lane 2: 0.5 µg Thyroid RNA
Detection	CDP-Star, ready-to-use
Exposure time	10 min

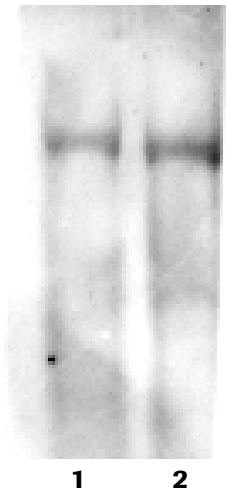
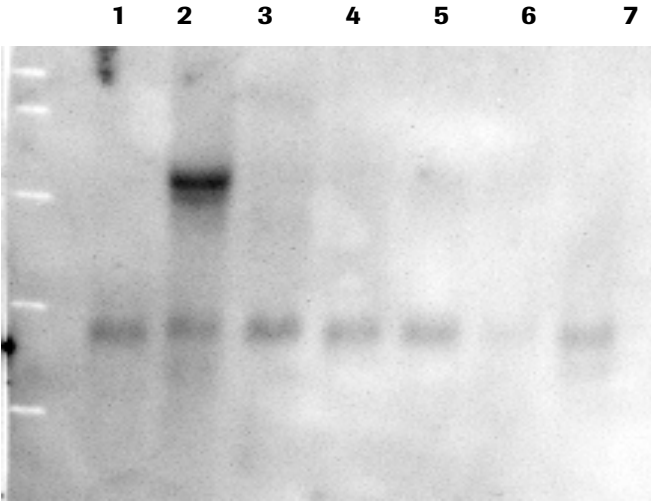


Fig. 3

Different tissues, normalized to Actin, hybridized with TSH-receptor probe (100 ng/ml) in a Northern Blot.

Probe	DIG-labeled TSH receptor probe (100 ng/ml) and DIG-labeled Actin probe (100 ng/ml).	
Template	Lane	
	1	Stomach
	2	Thyroid
	3	Spinal cord
	4	Lymph node
	5	Trachea
	6	Adrenal gland
	7	Bone marrow



5. Appendix

5.1 Trouble shooting

Trouble shooting table This table describes various troubleshooting parameters for DIG-labeling and detection

Problem	Possible cause	Recommendation
Low sensitivity	Inefficient probe labeling	• Check labeling efficiency (section 3.5). • Purify template DNA by phenolization and ethanol precipitation • Make sure that the template was linearized before labeling. • Do not store template in buffers containing more than 0.1 mM EDTA. • Check the amount and quality of target RNA
	Low probe concentration in the hybridization	• Increase probe concentration to 100 ng/ml . • Prolong hybridization time to overnight.
High background	Concentration of labeled probe is too high	Determine optimal probe concentration as described in section 3.5, do not use more than 100 ng/ml.
	Drying of the membrane	Make sure the membrane did not dry out at any time.
	Probe sequence	Make sure that the probe does not contain cross-hybridizing vector sequences.
	Inefficient prehybridization	Make sure that the membrane was soaked in sufficient prehybridization solution.
	Wrong type of nylon membrane	Some types of nylon membrane may cause high background: use nylon membranes from Roche Molecular Biochemicals, tested for the DIG-System.
	Inefficient stringency washes	• Check temperature of stringency washes, prewarm wash solution to correct temperature. • Use 0.1 x SSC for high stringency wash

Problem	Possible cause	Recommendation
Smear in lanes	Target concentration too high	Do not use more than 1 µg of total or 100 ng of mRNA per lane. The DIG System is more sensitive than radioactivity and higher RNA concentrations result in detection of degradation products.
Special hints for immunoassay		• Make sure, that you use only RNase free equipment for the whole detection procedure. • Do not allow the membrane to dry during detection procedure. • Handle membranes only with powder-free gloves. • Use sufficient volumes of all solutions. • Make sure that membranes do not stick to trays during detection. Avoid that membranes stick together. • Shake membranes during the whole detection procedure. • When using laboratory trays for the detection procedure, they should be rigorously cleaned before use. Anti-DIG-AP binding and chemiluminescent development should be done in separate trays. • Work under clean conditions when handling the chemiluminescent substrate solution, and avoid phosphatase contamination. • For chemiluminescent detection membranes should also be sealed in plastic bags. • Do not use "Saran Wrap".

5.2 Changes to previous version

Editorial corrections.
Update trademark information.
Deleted license disclaimer.

Regulatory Disclaimer

For life science research only. Not for use in diagnostic procedures.

Trademarks

DIG EASY HYB, EXPAND, and HIGH PURE are trademarks of Roche.
CDP-Star is a trademark of Applera Corporation.
Other brands or product names are trademarks of their respective holders.

5.3

References

- 1 Høltke, H.J. et al. (1995) The Digoxigenin (DIG) System for non-radioactive labeling and detection of nucleic acids-an overview. *Cell. Mol. Biol.* **41**: 883.
- 2 Sambrook, J., Fritsch, E.M. and Maniatis, T. (1989) Molecular Cloning: A Laboratory Manual, 2nd Edition, Cold Spring Harbor Laboratory, Cold Spring Harbor Labor, New York.
- 3 Logel, J.; Dill, D. & Leonard, S. (1992) Synthesis of cRNA probes from PCR-generated DNA, *Biotechniques* **13**, 604-610. Please refer to our website for the following informations:
- 4 DIG Product Selection Guide
- 5 DIG Application Manual for Filter Hybridization
- 6 Non-radioactive In situ Hybridization Manual
- 7 Lab FAQs

5.4

Ordering Information

Kits

For a complete overview, please visit and bookmark our "DIG Reagents and Kits for Non-Radioactive Nucleic Acid Labeling and Detection" Special Interest Site at <http://www.roche-applied-science.com/DIG>

Product	Pack Size	Cat. No
DIG Wash and Block Buffer Set	30 blots (10 × 10 cm ²)	11 585 762 001
Expand High Fidelity PCR System	100 U	11 732 641 001
High Pure PCR Product Purification Kit	50 purifications 250 purifications	11 732 668 001 11 732 676 001
High Pure Plasmid Isolation Kit	50 purifications 250 purifications	11 754 777 001 11 754 785 001

Single reagents

Product	Pack Size	Cat. No.
Agarose MP	20 g	11 444 964 001
DIG Easy Hyb (ready-to-use hybridization solution without formamide)	500 ml	11 603 558 001
DIG Easy Hyb Granules	1 set (6 x 100 ml)	11 796 895 001
DIG Wash and Block Buffer Set	1 set	11 585 762 001
Hybridization bags	50 bags	11 666 649 001
Nylon Membrane, positively charged (20 × 30 cm) (10 × 15 cm) (0.3 × 3 m roll)	10 sheets	11 209 272 001
	20 sheets	11 209 299 001
	1 roll	11 417 240 001
RNA Molecular Weight Marker, digoxigenin-labeled:		
RNA Molecular Weight Marker I	4 µg (200 µl)	11 526 529 910
RNA Molecular Weight Marker II	2 µg (200 µl)	11 526 537 910
RNA Molecular Weight Marker III	2 µg (200 µl)	11 373 099 910
SSC, Buffers in a box, 20x	4 l	11 666 681 001

* available from Roche Applied Science

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